

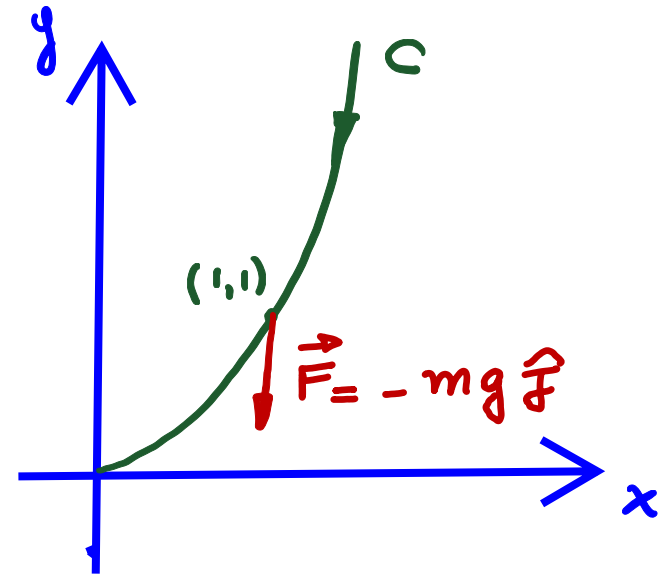
Midterm Exam Problem 4 Fall 2016

A bead of mass " m " slides without friction down a wire bent in the shape of the curve $y = x^2$ under the influence of the gravitational force $-mg\hat{j}$.

The speed of the bead as it passes through the point $(1,1)$ is " V ". At that instant, find

a. The normal acceleration of the bead in terms of " V "

a. The rate of change of its speed in terms of " g ".



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Tangential and normal components of acceleration

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Q) TO CALCULATE $\underbrace{Q_N}$ WE NEED TO PARAMETRIZE THE CURVE

$$\begin{aligned} x &= t \\ y &= t^2 \end{aligned}$$

WRONG
PARAMETRIZATION

$$\begin{aligned} x &= -t \\ y &= (-t)^2 = t^2 \end{aligned} \quad ; \quad t \leq 0$$

$$\begin{aligned} t = -1 &\rightarrow (1, 1) \\ t = 0 &\rightarrow (0, 0) \end{aligned}$$

GOING DOWN
THE WIRE ✓

THE VECTOR FUNCTION IS GIVEN BY,

$$\vec{r}(t) = -t\hat{i} + t^2\hat{j} \quad ; \quad t \leq 0$$



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NORMAL ACC ;

$$a_N = \kappa |\vec{v}|^2, \quad \kappa = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3}$$

PHYSICAL SPEED

GEOMETRIC CURVATURE

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = -\hat{i} + 2t\hat{j}$$

$$|\vec{v}(t)| = \sqrt{1 + 4t^2}$$

$$\text{AT POINT } (1,1) \Rightarrow t = -1 \Rightarrow |\vec{v}(t)|_{t=-1} = \sqrt{5}$$

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = 2\hat{j}$$



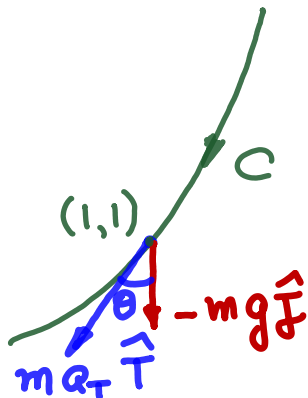
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$$|\vec{v} \times \vec{a}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2t & 0 \\ 0 & 2 & 0 \end{vmatrix} = |-2\hat{k}| = 2$$

CURVATURE AT $(1,1)$: $K = \frac{2}{(\sqrt{5})^3}$, $|\vec{v}| = V$

$\Rightarrow Q_N = K |\vec{v}|^2 = \frac{2}{5\sqrt{5}} V^2$ ← PHYSICAL SPEED AT $(1,1)$

b) RATE OF CHANGE OF SPEED



BY SECOND NEWTON'S LAW, ALONG THE TANGENTIAL COMPONENT OF ACC.

$$m a_T = -m g \hat{j} \cdot \hat{T}$$



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$$a_T = -g \hat{f} \cdot \hat{T}$$

$$\hat{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{-i + 2t \hat{f}}{|\vec{v}|} = \frac{-i + 2t \hat{f}}{\sqrt{5}}$$

$$\hat{T}_{(1,1)} = \frac{-i + 2\hat{f}}{\sqrt{5}}$$

$$a_T|_{(1,1)} = -g \hat{f} \cdot \frac{(-i - 2\hat{f})}{\sqrt{5}} = \frac{2g}{\sqrt{5}}$$

WE CAN WRITE THE ACCELERATION AT (1,1) AS,

$$\vec{a}|_{(1,1)} = \frac{2g}{\sqrt{5}} \hat{T} + \frac{2v^2}{5\sqrt{5}} \hat{N}$$

